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Authentication Routes

Base Path: `/api/v1/auth`

All authentication routes are public except `/auth/me` and `/auth/logout` which require authentication.

1. Register New User

POST `/api/v1/auth/register`
Content-Type: `application/json`

Description: Register a new user account. Account is inactive until approved by SUPERADMIN.

Request Body:

```
{
  "employeeId": "EMP001",
  "name": "John Doe",
  "email": "john.doe@railway.com",
  "phone": "+91-9876543210",
  "password": "SecurePass123!",
  "division": "Operations",
  "designation": "Shift Coordinator",
  "role": "ADMIN"
}
```

Validation Rules:

- `employeeId` : Required, unique, string
- `name` : Required, min 2 characters
- `email` : Required, valid email format, unique
- `phone` : Optional, string
- `password` : Required, min 6 characters
- `division` : Optional, string

- `designation` : Optional, string
- `role` : Optional, enum: `USER` , `ADMIN` , `SUPERADMIN` (default: `USER`)

Success Response (201 Created):

```
{
  "success": true,
  "message": "Registration successful. Your account is pending approval by administrator.",
  "data": {
    "user": {
      "id": "550e8400-e29b-41d4-a716-446655440000",
      "employeeId": "EMP001",
      "name": "John Doe",
      "email": "john.doe@railway.com",
      "phone": "+91-9876543210",
      "role": "ADMIN",
      "status": "INACTIVE",
      "isVerified": false,
      "division": "Operations",
      "designation": "Shift Coordinator",
      "createdAt": "2026-03-25T10:30:00.000Z"
    }
  }
}
```

Error Response (400 Bad Request):

```
{
  "success": false,
  "message": "Validation failed",
  "errors": [
    {
      "field": "email",
      "message": "Email already exists"
    }
  ]
}
```

Rate Limit: 5 requests per 15 minutes per IP

2. Login

POST `/api/v1/auth/login`
Content-Type: `application/json`

Description: Authenticate user and receive JWT tokens.

Request Body:

```
{
  "email": "john.doe@railway.com",
  "password": "SecurePass123!"
}
```

Validation Rules:

- `email` : Required, valid email format
- `password` : Required, non-empty string

Success Response (200 OK):

```
{
  "success": true,
  "message": "Login successful",
  "data": {
    "user": {
      "id": "550e8400-e29b-41d4-a716-446655440000",
      "employeeId": "EMP001",
      "name": "John Doe",
      "email": "john.doe@railway.com",
      "phone": "+91-9876543210",
      "role": "ADMIN",
      "status": "ACTIVE",
      "division": "Operations",
      "createdAt": "2025-11-24T13:30:00.000Z",
      "lastLogin": "2026-03-25T10:35:00.000Z"
    },
    "tokens": {
      "accessToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
      "refreshToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
    }
  }
}
```

Error Response (401 Unauthorized):

```
{
  "success": false,
  "message": "Invalid email or password"
}
```

Error Response (403 Forbidden):

```
{
  "success": false,
```

```
"message": "Account is inactive or suspended"
}
```

Rate Limit: 10 requests per 15 minutes per IP

3. Get Current User

```
GET /api/v1/auth/me
Authorization: Bearer <accessToken>
```

Description: Retrieve details of the currently authenticated user.

Success Response (200 OK):

```
{
  "success": true,
  "data": {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "employeeid": "EMP001",
    "name": "John Doe",
    "email": "john.doe@railway.com",
    "phone": "+91-9876543210",
    "role": "ADMIN",
    "status": "ACTIVE",
    "division": "Operations",
    "designation": "Shift Coordinator",
    "priority": 1,
    "isVerified": true,
    "verifiedAt": "2025-11-27T10:00:00.000Z",
    "createdAt": "2025-11-24T13:30:00.000Z",
    "lastLogin": "2026-03-25T10:35:00.000Z"
  }
}
```

Error Response (401 Unauthorized):

```
{
  "success": false,
  "message": "No token provided or invalid token"
}
```

4. Refresh Access Token

```
POST /api/v1/auth/refresh
```

Content-Type: application/json

Description: Generate a new access token using refresh token when access token expires.

Request Body:

```
{
  "refreshToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
}
```

Validation Rules:

- `refreshToken` : Required, non-empty string

Success Response (200 OK):

```
{
  "success": true,
  "message": "Token refreshed successfully",
  "data": {
    "accessToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
    "refreshToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
  }
}
```

Error Response (401 Unauthorized):

```
{
  "success": false,
  "message": "Invalid or expired refresh token"
}
```

5. Logout

POST /api/v1/auth/logout
Authorization: Bearer <accessToken>

Description: Logout the current user. Client should delete stored tokens.

Success Response (200 OK):

```
{
  "success": true,
```

```
"message": "Logout successful"
}
```

6. Forgot Password

```
POST /api/v1/auth/forgot-password
Content-Type: application/json
```

Description: Request password reset OTP. OTP will be sent to the registered email.

Request Body:

```
{
  "email": "john.doe@railway.com"
}
```

Success Response (200 OK):

```
{
  "success": true,
  "message": "OTP sent to your registered email"
}
```

Error Response (404 Not Found):

```
{
  "success": false,
  "message": "User not found"
}
```

Rate Limit: 3 requests per 15 minutes per IP

7. Reset Password

```
POST /api/v1/auth/reset-password
Content-Type: application/json
```

Description: Reset password using OTP received via email.

Request Body:

```
{
  "email": "john.doe@railway.com",
  "otp": "123456",
  "newPassword": "NewSecurePass123!"
}
```

Validation Rules:

- email : Required, valid email format
- otp : Required, 6 digits
- newPassword : Required, min 6 characters

Success Response (200 OK):

```
{
  "success": true,
  "message": "Password reset successful"
}
```

Shift Management Routes

Base Path: /api/v1/shifts

Authentication Required: All routes require Bearer token

1. Create New Shift

POST /api/v1/shifts
Authorization: Bearer <accessToken>
Content-Type: application/json

Permission: ADMIN, SUPERADMIN

Description: Create a new shift and start duty hour tracking.

Request Body:

```
{
  "trainNumber": "12345",
  "trainName": "Rajdhani Express",
  "locomotiveNo": "WAP-7-30456",
  "locoPilot": {
    "employeeId": "LP001",
    "name": "Rajesh Kumar",
    "phone": "+91-9876543210"
  }
}
```

```
{
  "trainManager": {
    "employeeId": "TM001",
    "name": "Suresh Singh",
    "phone": "+91-9876543211"
  },
  "trainArrivalDateTime": "2026-03-25T08:30:00.000Z",
  "signOnDateTime": "2026-03-25T08:00:00.000Z",
  "timeOfTO": "2026-03-25T08:45:00.000Z",
  "departureDateTime": "2026-03-25T09:00:00.000Z",
  "signOnStation": "NDLS",
  "section": "Delhi-Mumbai",
  "dutyType": "SP"
}
```

Validation Rules:

- trainNumber : Required, string
- trainName : Optional, string
- locomotiveNo : Required, string
- locoPilot.employeeId : Required, string
- locoPilot.name : Required, string
- locoPilot.phone : Optional, valid phone number
- trainManager.employeeId : Required, string
- trainManager.name : Required, string
- trainManager.phone : Optional, valid phone number
- trainArrivalDateTime : Required, ISO 8601 datetime
- signOnDateTime : Required, ISO 8601 datetime
- timeOfTO : Optional, ISO 8601 datetime
- departureDateTime : Optional, ISO 8601 datetime
- signOnStation : Required, string
- section : Required, string
- dutyType : Required, enum: SP , WR , LR

Success Response (201 Created):

```
{
  "success": true,
  "message": "Shift created successfully",
  "data": {
    "id": "shift-550e8400-e29b-41d4-a716-446655440000",
    "trainNumber": "12345",
    "trainName": "Rajdhani Express",
    "locomotiveNo": "WAP-7-30456",
    "locomotiveId": "loco-550e8400",
    "status": "IN_PROGRESS",
    "dutyType": "SP",
  }
}
```

```
"section": "Delhi-Mumbai",
"trainArrivalDateTime": "2026-03-25T08:30:00.000Z",
"signOnDateTime": "2026-03-25T08:00:00.000Z",
"timeOfTO": "2026-03-25T08:45:00.000Z",
"departureDateTime": "2026-03-25T09:00:00.000Z",
"signOnStation": "NDLS",
"dutyHours": null,
"reliefRequired": false,
"reliefPlanned": false,
"locoPilot": {
  "id": "pilot-550e8400",
  "employeeId": "LP001",
  "name": "Rajesh Kumar",
  "phone": "+91-9876543210",
  "status": "ON_DUTY"
},
"trainManager": {
  "id": "manager-550e8400",
  "employeeId": "TM001",
  "name": "Suresh Singh",
  "phone": "+91-9876543211",
  "status": "ON_DUTY"
},
"createdAt": "2026-03-25T08:00:00.000Z",
"updatedAt": "2026-03-25T08:00:00.000Z"
}
```

Rate Limit: 20 requests per hour per user

2. List All Shifts

```
GET /api/v1/shifts?status=IN_PROGRESS&page=1&limit=10
Authorization: Bearer <accessToken>
```

Permission: All authenticated users

Description: Retrieve all shifts with optional filtering and pagination.

Query Parameters:

- `status` : Optional, enum: `SCHEDULED` , `IN_PROGRESS` , `COMPLETED` , `RELIEF_PLANNED` , `CANCELLED`
- `trainNumber` : Optional, string
- `section` : Optional, string
- `dutyType` : Optional, enum: `SP` , `WR` , `LR`
- `fromDate` : Optional, ISO 8601 date
- `toDate` : Optional, ISO 8601 date
- `page` : Optional, default: 1

- `limit` : Optional, default: 10, max: 100

Success Response (200 OK):

```
{
  "success": true,
  "data": {
    "shifts": [
      {
        "id": "shift-550e8400-e29b-41d4-a716-446655440000",
        "trainNumber": "12345",
        "trainName": "Rajdhani Express",
        "locomotiveNo": "WAP-7-30456",
        "status": "IN_PROGRESS",
        "section": "Delhi-Mumbai",
        "dutyType": "SP",
        "trainArrivalDateTime": "2026-03-25T08:30:00.000Z",
        "signOnDateTime": "2026-03-25T08:00:00.000Z",
        "currentDutyHours": 5.75,
        "reliefRequired": false,
        "reliefPlanned": false,
        "locoPilot": {
          "id": "pilot-550e8400",
          "employeeId": "LP001",
          "name": "Rajesh Kumar"
        },
        "trainManager": {
          "id": "manager-550e8400",
          "employeeId": "TM001",
          "name": "Suresh Singh"
        },
        "createdAt": "2026-03-25T08:00:00.000Z"
      }
    ],
    "pagination": {
      "page": 1,
      "limit": 10,
      "total": 50,
      "totalPages": 5
    }
  }
}
```

3. Get Shift by ID

```
GET /api/v1/shifts/:id
Authorization: Bearer <accessToken>
```

Permission: All authenticated users

Description: Retrieve detailed information about a specific shift.

Path Parameters:

- `id` : Required, UUID of the shift

Success Response (200 OK):

```
{
  "success": true,
  "data": {
    "id": "shift-550e8400-e29b-41d4-a716-446655440000",
    "trainNumber": "12345",
    "trainName": "Rajdhani Express",
    "locomotiveNo": "WAP-7-30456",
    "locomotiveId": "loco-550e8400",
    "status": "IN_PROGRESS",
    "dutyType": "SP",
    "section": "Delhi-Mumbai",
    "trainArrivalDateTime": "2026-03-25T08:30:00.000Z",
    "signOnDateTime": "2026-03-25T08:00:00.000Z",
    "timeOfTO": "2026-03-25T08:45:00.000Z",
    "departureDateTime": "2026-03-25T09:00:00.000Z",
    "signOffDateTime": null,
    "signOnStation": "NDLS",
    "signOffStation": null,
    "dutyHours": null,
    "reliefRequired": false,
    "reliefPlanned": false,
    "reliefTime": null,
    "reliefReason": null,
    "alert7HrSent": false,
    "alert8HrSent": true,
    "alert8HrResponse": "PLAN_RELIEF",
    "alert9HrSent": true,
    "alert9HrResponse": "CREW_RELIEVED",
    "locoPilot": {
      "id": "pilot-550e8400",
      "employeeId": "LP001",
      "name": "Rajesh Kumar",
      "phone": "+91-9876543210",
      "status": "ON_DUTY"
    },
    "trainManager": {
      "id": "manager-550e8400",
      "employeeId": "TM001",
      "name": "Suresh Singh",
      "phone": "+91-9876543211",
      "status": "ON_DUTY"
    },
    "createdAt": "2026-03-25T08:00:00.000Z",
    "updatedAt": "2026-03-25T10:00:00.000Z"
  }
}
```

```
}  
}
```

Error Response (404 Not Found):

```
{  
  "success": false,  
  "message": "Shift not found"  
}
```

4. Update Shift

PATCH /api/v1/shifts/:id
Authorization: Bearer <accessToken>
Content-Type: application/json

Permission: ADMIN, SUPERADMIN

Description: Update shift details (all fields optional).

Request Body (example):

```
{  
  "timeOfTO": "2026-03-25T08:50:00.000Z",  
  "departureDateTime": "2026-03-25T09:10:00.000Z",  
  "section": "Delhi-Mumbai-Bangalore",  
  "dutyType": "WR",  
  "reliefPlanned": true,  
  "reliefReason": "Crew rotation"  
}
```

Success Response (200 OK):

```
{  
  "success": true,  
  "message": "Shift updated successfully",  
  "data": {  
    "id": "shift-550e8400-e29b-41d4-a716-446655440000",  
    "trainNumber": "12345",  
    "status": "IN_PROGRESS",  
    "section": "Delhi-Mumbai-Bangalore",  
    "dutyType": "WR",  
    "timeOfTO": "2026-03-25T08:50:00.000Z",  
    "departureDateTime": "2026-03-25T09:10:00.000Z",  
    "reliefPlanned": true,  
    "reliefReason": "Crew rotation",  
    "updatedAt": "2026-03-25T10:05:00.000Z"  
  }  
}
```

```
}  
}
```

5. Complete Shift

```
POST /api/v1/shifts/:id/complete  
Authorization: Bearer <accessToken>  
Content-Type: application/json
```

Permission: ADMIN, SUPERADMIN

Description: Mark a shift as completed with sign-off details.

Request Body:

```
{  
  "signOffDateTime": "2026-03-25T16:30:00.000Z",  
  "signOffStation": "MUMBAI"  
}
```

Validation Rules:

- `signOffDateTime` : Required, ISO 8601 datetime
- `signOffStation` : Required, string

Success Response (200 OK):

```
{  
  "success": true,  
  "message": "Shift completed successfully",  
  "data": {  
    "id": "shift-550e8400-e29b-41d4-a716-446655440000",  
    "trainNumber": "12345",  
    "status": "COMPLETED",  
    "signOffDateTime": "2026-03-25T16:30:00.000Z",  
    "signOffStation": "MUMBAI",  
    "dutyHours": 8.5,  
    "updatedAt": "2026-03-25T16:30:00.000Z"  
  }  
}
```

6. Delete Shift

```
DELETE /api/v1/shifts/:id
Authorization: Bearer <accessToken>
```

Permission: SUPERADMIN only

Description: Delete a shift (only SUPERADMIN can delete).

Success Response (200 OK):

```
{
  "success": true,
  "message": "Shift deleted successfully"
}
```

Error Response (403 Forbidden):

```
{
  "success": false,
  "message": "You do not have permission to delete shifts"
}
```

7. Get Active Shifts Summary

```
GET /api/v1/shifts/active/summary
Authorization: Bearer <accessToken>
```

Permission: All authenticated users

Description: Get summary statistics of all active shifts.

Success Response (200 OK):

```
{
  "success": true,
  "data": {
    "totalActiveShifts": 15,
    "shiftsWithAlerts": 5,
    "shiftsNearingDutyLimit": 8,
    "averageDutyHours": 6.5,
    "dutyLimitExceeded": 2,
    "recentAlerts": [
      {
        "shiftId": "shift-550e8400",
        "trainNumber": "12345",
        "alert": "8HR",
        "time": "2026-03-25T16:00:00.000Z"
      }
    ]
  }
}
```

```
}  
]  
}  
}
```

8. Submit Alert Response

```
POST /api/v1/shifts/:id/alert-response  
Authorization: Bearer <accessToken>  
Content-Type: application/json
```

Permission: ADMIN, SUPERADMIN

Description: Submit response to duty hour alerts (8HR, 9HR, 10HR, 11HR, 14HR).

Request Body:

```
{  
  "alertType": "8HR",  
  "response": "PLAN_RELIEF",  
  "remarks": "Relief crew will arrive within 2 hours"  
}
```

Validation Rules:

- `alertType` : Required, enum: 8HR , 9HR , 10HR , 11HR , 14HR
- `response` : Required, string (varies by alert type)
- `remarks` : Optional, string

Success Response (200 OK):

```
{  
  "success": true,  
  "message": "Alert response recorded",  
  "data": {  
    "shiftId": "shift-550e8400",  
    "alertType": "8HR",  
    "response": "PLAN_RELIEF",  
    "recordedAt": "2026-03-25T16:05:00.000Z"  
  }  
}
```

9. Get Shift Alert History

```
GET /api/v1/shifts/:id/alerts
Authorization: Bearer <accessToken>
```

Permission: All authenticated users

Description: Retrieve complete alert history for a shift.

Success Response (200 OK):

```
{
  "success": true,
  "data": {
    "shiftId": "shift-550e8400",
    "trainNumber": "12345",
    "alerts": [
      {
        "alertType": "8HR",
        "sentAt": "2026-03-25T16:00:00.000Z",
        "response": "PLAN_RELIEF",
        "responseAt": "2026-03-25T16:05:00.000Z"
      },
      {
        "alertType": "9HR",
        "sentAt": "2026-03-25T17:00:00.000Z",
        "response": "CREW_RELIEVED",
        "responseAt": "2026-03-25T17:15:00.000Z"
      }
    ]
  }
}
```

Dashboard Routes

Base Path: `/api/v1/dashboard`

Authentication Required: All routes require Bearer token

1. Get Dashboard Statistics

```
GET /api/v1/dashboard/stats
Authorization: Bearer <accessToken>
```

Description: Get comprehensive dashboard statistics.

Success Response (200 OK):

```
{
  "success": true,
  "data": {
    "totalShifts": 150,
    "activeShifts": 12,
    "completedShifts": 135,
    "cancelledShifts": 3,
    "averageDutyHours": 7.8,
    "shiftsExceedingDutyLimit": 5,
    "pendingAlertResponses": 2,
    "reliefPlannedCount": 3,
    "topTrainNumbers": ["12345", "54321", "98765"],
    "topStations": ["NDLS", "MUMBAI", "BANGALORE"],
    "lastUpdated": "2026-03-25T10:30:00.000Z"
  }
}
```

2. Get Recent Activities

```
GET /api/v1/dashboard/recent-activities?limit=20&offset=0&type=SIGN_ON
Authorization: Bearer <accessToken>
```

Description: Get recent activities/duty logs with filtering.

Query Parameters:

- `limit` : Optional, default: 10, max: 100
- `offset` : Optional, default: 0
- `type` : Optional, filter by activity type (SIGN_ON, SIGN_OFF, ALERT_8HR, ALERT_9HR, etc.)

Valid Activity Types:

- SIGN_ON, SIGN_OFF
- ALERT_7HR, ALERT_8HR, ALERT_9HR, ALERT_10HR, ALERT_11HR, ALERT_14HR
- RELIEF_PLANNED, RELIEF_NOT_REQUIRED, CREW_RELIEVED, CREW_NOT_BOOKED
- KEEP_ON_DUTY, CREW_ALREADY_RELIEVED, RELEASE

Success Response (200 OK):

```
{
  "success": true,
  "data": {
    "activities": [
      {
        "id": "log-550e8400",
        "shiftId": "shift-550e8400",
        "trainNumber": "12345",

```

```
"staffName": "Rajesh Kumar",
"staffType": "LOCO_PILOT",
"logType": "SIGN_ON",
"logTime": "2026-03-25T08:00:00.000Z",
"dutyHoursAtLog": 0,
"remarks": null
},
{
  "id": "log-550e8401",
  "shiftId": "shift-550e8400",
  "trainNumber": "12345",
  "staffName": "Rajesh Kumar",
  "staffType": "LOCO_PILOT",
  "logType": "ALERT_8HR",
  "logTime": "2026-03-25T16:00:00.000Z",
  "dutyHoursAtLog": 8,
  "remarks": "Relief crew informed"
}
],
"pagination": {
  "offset": 0,
  "limit": 20,
  "total": 150
}
}
```

3. Get Shift Trends

```
GET /api/v1/dashboard/trends?days=30
Authorization: Bearer <accessToken>
```

Description: Get shift trends for charts over last N days.

Query Parameters:

- `days` : Optional, default: 7, max: 90

Success Response (200 OK):

```
{
  "success": true,
  "data": {
    "trends": [
      {
        "date": "2026-02-24",
        "shiftsCreated": 12,
        "shiftsCompleted": 10,
        "averageDutyHours": 8.1,
        "shiftsExceedingLimit": 2,

```

```
"reliefRequired": 3
},
{
  "date": "2026-02-25",
  "shiftsCreated": 15,
  "shiftsCompleted": 14,
  "averageDutyHours": 7.9,
  "shiftsExceedingLimit": 1,
  "reliefRequired": 2
}
],
"summary": {
  "periodDays": 30,
  "totalShifts": 425,
  "averageDailyShifts": 14.17,
  "peakDay": "2026-03-15",
  "peakShifts": 18
}
}
}
```

4. Get Alerts Summary

```
GET /api/v1/dashboard/alerts-summary
Authorization: Bearer <accessToken>
```

Description: Get summary of active and pending alerts.

Success Response (200 OK):

```
{
  "success": true,
  "data": {
    "activeAlerts": {
      "alert8Hr": 3,
      "alert9Hr": 2,
      "alert10Hr": 1,
      "alert11Hr": 0,
      "alert14Hr": 0
    },
    "pendingResponses": 4,
    "recentAlerts": [
      {
        "shiftId": "shift-550e8400",
        "trainNumber": "12345",
        "alert": "8HR",
        "sentAt": "2026-03-25T16:00:00.000Z",
        "pendingSince": "30 minutes",
        "locoPilot": "Rajesh Kumar"
      }
    ]
  }
}
```

```
],  
  "reliefPlanned": 2,  
  "reliefNotRequired": 1,  
  "lastUpdated": "2026-03-25T10:30:00.000Z"  
}  
}
```

User Management Routes

Base Path: `/api/v1/users`

Authentication Required: All routes require Bearer token + SUPERADMIN role

1. Get All Users

GET `/api/v1/users?status=ACTIVE&role=ADMIN&page=1&limit=10`
Authorization: Bearer `<accessToken>`

Permission: SUPERADMIN only

Description: Get all users with optional filtering.

Query Parameters:

- `status` : Optional, enum: `ACTIVE` , `INACTIVE` , `SUSPENDED`
- `role` : Optional, enum: `USER` , `ADMIN` , `SUPERADMIN`
- `division` : Optional, string
- `page` : Optional, default: 1
- `limit` : Optional, default: 10, max: 100

Success Response (200 OK):

```
{  
  "success": true,  
  "data": {  
    "users": [  
      {  
        "id": "550e8400-e29b-41d4-a716-446655440000",  
        "employeeId": "EMP001",  
        "name": "John Doe",  
        "email": "john.doe@railway.com",  
        "phone": "+91-9876543210",  
        "role": "ADMIN",  
        "status": "ACTIVE",  
        "division": "Operations",  
        "designation": "Shift Coordinator",  
        "priority": 1,  
      }  
    ]  
  }  
}
```

```
"isVerified": true,
"verifiedAt": "2025-11-27T10:00:00.000Z",
"createdAt": "2025-11-24T13:30:00.000Z",
"lastLogin": "2026-03-25T10:35:00.000Z"
}
],
"pagination": {
  "page": 1,
  "limit": 10,
  "total": 45,
  "totalPages": 5
}
}
}
```

2. Get Pending User Requests

```
GET /api/v1/users/pending-requests
Authorization: Bearer <accessToken>
```

Permission: SUPERADMIN only

Description: Get list of pending user registration/approval requests.

Success Response (200 OK):

```
{
  "success": true,
  "data": {
    "pendingRequests": [
      {
        "id": "550e8400-e29b-41d4-a716-446655440000",
        "employeeId": "EMP002",
        "name": "Jane Smith",
        "email": "jane.smith@railway.com",
        "phone": "+91-9876543211",
        "role": "USER",
        "division": "Traffic",
        "designation": "Station Master",
        "requestedAt": "2026-03-24T15:30:00.000Z",
        "status": "INACTIVE"
      }
    ],
    "totalPending": 3
  }
}
```

3. Get User by ID

```
GET /api/v1/users/:id
Authorization: Bearer <accessToken>
```

Permission: SUPERADMIN only

Description: Get detailed information of a specific user.

Path Parameters:

- `id` : Required, UUID of the user

Success Response (200 OK):

```
{
  "success": true,
  "data": {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "employeeId": "EMP001",
    "name": "John Doe",
    "email": "john.doe@railway.com",
    "phone": "+91-9876543210",
    "role": "ADMIN",
    "status": "ACTIVE",
    "division": "Operations",
    "designation": "Shift Coordinator",
    "priority": 1,
    "isVerified": true,
    "verifiedAt": "2025-11-27T10:00:00.000Z",
    "verifiedBy": "SUPERADMIN001",
    "createdAt": "2025-11-24T13:30:00.000Z",
    "lastLogin": "2026-03-25T10:35:00.000Z"
  }
}
```

4. Approve User

```
POST /api/v1/users/:id/approve
Authorization: Bearer <accessToken>
Content-Type: application/json
```

Permission: SUPERADMIN only

Description: Approve a pending user registration.

Request Body:

```
{}
```

Success Response (200 OK):

```
{
  "success": true,
  "message": "User approved successfully",
  "data": {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "employeeId": "EMP001",
    "name": "John Doe",
    "email": "john.doe@railway.com",
    "status": "ACTIVE",
    "isVerified": true,
    "verifiedAt": "2026-03-25T10:45:00.000Z"
  }
}
```

5. Reject User

```
POST /api/v1/users/:id/reject
Authorization: Bearer <accessToken>
Content-Type: application/json
```

Permission: SUPERADMIN only

Description: Reject a pending user registration.

Request Body:

```
{
  "reason": "Invalid employee ID"
}
```

Validation Rules:

- `reason` : Optional, string

Success Response (200 OK):

```
{
  "success": true,
  "message": "User rejected successfully",
  "data": {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "employeeId": "EMP001",
  }
}
```

```
"status": "INACTIVE",
"rejectedAt": "2026-03-25T10:50:00.000Z",
"rejectionReason": "Invalid employee ID"
}
}
```

6. Change User Role

```
PATCH /api/v1/users/:id/role
Authorization: Bearer <accessToken>
Content-Type: application/json
```

Permission: SUPERADMIN only

Description: Change user role (USER, ADMIN, SUPERADMIN).

Request Body:

```
{
  "role": "ADMIN"
}
```

Validation Rules:

- `role` : Required, enum: `USER` , `ADMIN` , `SUPERADMIN`

Success Response (200 OK):

```
{
  "success": true,
  "message": "User role updated",
  "data": {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "employeeId": "EMP001",
    "name": "John Doe",
    "role": "ADMIN",
    "updatedAt": "2026-03-25T10:55:00.000Z"
  }
}
```

7. Update User

```
PATCH /api/v1/users/:id
Authorization: Bearer <accessToken>
```

Content-Type: application/json

Permission: SUPERADMIN only

Description: Update user details (all fields optional).

Request Body (example):

```
{
  "phone": "+91-9876543210",
  "division": "Traffic",
  "designation": "Traffic Inspector",
  "priority": 2
}
```

Success Response (200 OK):

```
{
  "success": true,
  "message": "User updated successfully",
  "data": {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "phone": "+91-9876543210",
    "division": "Traffic",
    "designation": "Traffic Inspector",
    "priority": 2,
    "updatedAt": "2026-03-25T11:00:00.000Z"
  }
}
```

8. Activate User

POST /api/v1/users/:id/activate
Authorization: Bearer <accessToken>
Content-Type: application/json

Permission: SUPERADMIN only

Description: Activate a deactivated user.

Success Response (200 OK):

```
{
  "success": true,
  "message": "User activated",
  "data": {
    "id": "550e8400-e29b-41d4-a716-446655440000",

```

```
"employeeId": "EMP001",
"status": "ACTIVE",
"updatedAt": "2026-03-25T11:05:00.000Z"
}
}
```

9. Deactivate User

```
POST /api/v1/users/:id/deactivate
Authorization: Bearer <accessToken>
Content-Type: application/json
```

Permission: SUPERADMIN only

Description: Deactivate an active user.

Success Response (200 OK):

```
{
  "success": true,
  "message": "User deactivated",
  "data": {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "employeeId": "EMP001",
    "status": "INACTIVE",
    "updatedAt": "2026-03-25T11:10:00.000Z"
  }
}
```

10. Delete User

```
DELETE /api/v1/users/:id
Authorization: Bearer <accessToken>
```

Permission: SUPERADMIN only

Description: Permanently delete a user account.

Success Response (200 OK):

```
{
  "success": true,
  "message": "User deleted successfully"
}
```

Common Response Formats

Success Response Template

```
{
  "success": true,
  "message": "Operation successful",
  "data": {
    // Response-specific data
  }
}
```

Error Response Template

```
{
  "success": false,
  "message": "Error message",
  "errors": [
    {
      "field": "fieldName",
      "message": "Validation error message"
    }
  ]
}
```

Paginated Response Template

```
{
  "success": true,
  "data": {
    "items": [],
    "pagination": {
      "page": 1,
      "limit": 10,
      "total": 50,
      "totalPages": 5
    }
  }
}
```

Error Handling

HTTP Status Codes

Code	Meaning	Example
200	OK	Request successful
201	Created	Resource created successfully
400	Bad Request	Validation failed
401	Unauthorized	No token or invalid token
403	Forbidden	No permission for this action
404	Not Found	Resource doesn't exist
409	Conflict	Resource already exists
422	Unprocessable Entity	Business logic error
429	Too Many Requests	Rate limit exceeded
500	Server Error	Internal server error

Common Error Codes

```
// Invalid Token
{
  "success": false,
  "message": "No token provided or invalid token",
  "statusCode": 401
}
```

```
// Permission Denied
{
  "success": false,
  "message": "You do not have permission to perform this action",
  "statusCode": 403
}
```

```
// Resource Not Found
{
  "success": false,
  "message": "Shift not found",
  "statusCode": 404
}
```

```
// Validation Error
{
  "success": false,
  "message": "Validation failed",
}
```

```
"errors": [  
  {  
    "field": "trainNumber",  
    "message": "Train number is required"  
  }  
],  
"statusCode": 400  
}
```

```
// Rate Limit Exceeded  
{  
  "success": false,  
  "message": "Too many requests, please try again later",  
  "statusCode": 429  
}
```

Business Logic Errors

```
// Shift Already Completed  
{  
  "success": false,  
  "message": "Cannot update a completed shift",  
  "statusCode": 422  
}
```

```
// Invalid Time Range  
{  
  "success": false,  
  "message": "Sign-on time cannot be before train arrival time",  
  "statusCode": 422  
}
```

```
// Insufficient Permissions  
{  
  "success": false,  
  "message": "Only SUPERADMIN can delete shifts",  
  "statusCode": 403  
}
```

Authentication Header Format

All protected endpoints require the `Authorization` header:

Authorization: Bearer <access_token>

Example:

GET /api/v1/shifts HTTP/1.1

Host: api.railway.com

Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6IjU1MGU4NDAw...

Notes

- All timestamps are in ISO 8601 format (UTC)
 - All IDs are UUIDs (v4)
 - Rate limiting is applied to prevent abuse
 - Token expiry: Access token valid for 1 hour, Refresh token for 7 days
 - Pagination: Default page size is 10, max 100
-

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